

An Analysis of Logistic regression

Logistic regression

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$P(y = 1 | x) = \sigma(w^T x + b)$$

Logistic regression -- Part 1

This exponential relationship provides an interpretation for β_1 : The odds multiply by e^{β_1} for every 1-unit increase in x . For a binary independent variable the odds ratio is defined as $\frac{ad}{bc}$ where a, b, c and d are cells in a 2×2 contingency table.

Logistic regression -- Part 2

A group of 20 students spends between 0 and 6 hours studying for an exam. How does the number of hours spent studying affect the probability of the student passing the exam?

Logistic regression -- Part 3

The log loss can be interpreted as the "surprisal" of the actual outcome y_k relative to the prediction p_k , and is a measure of information content. Log loss is always greater than or equal to 0, equals 0 only in case of a perfect prediction (i.e., when $p_k = 1$ and $y_k = 1$, or $p_k = 0$ and $y_k = 0$), and approaches infinity as the prediction gets worse (i.e., when $y_k = 1$ and $p_k \rightarrow 0$ or $y_k = 0$ and $p_k \rightarrow 1$), meaning the actual outcome is "more surprising". Since the value of the logistic function is always strictly between zero and one, the log loss is always greater than zero and less than infinity. Unlike in a linear regression, where the model can have zero loss at a point by passing through a data point (and zero loss overall if all points are on a line), in a logistic regression it is not possible to have zero loss at any points, since y_k is either 0 or 1, but $0 < p_k < 1$. These can be combined into a single expression:

Logistic regression -- Part 4

The regression coefficients $\beta_0, \beta_1, \dots, \beta_m$ are grouped into a single vector β of size $m + 1$. For each data point i , an additional explanatory pseudo-variable $x_{0,i}$ is added, with a fixed value of

1, corresponding to the intercept coefficient β_0 . The resulting explanatory variables $x_{0,i}, x_{1,i}, \dots, x_{m,i}$ are then grouped into a single vector X_i of size $m + 1$. This makes it possible to write the linear predictor function as follows:

Logistic regression -- Part 5

$$\begin{aligned} \Pr(Y_i = 1 | X_i) &= \Pr(\epsilon_i > 0 | X_i) = \Pr(\beta \cdot X_i + \epsilon_i > 0) \\ &= \Pr(\epsilon_i > -\beta \cdot X_i) = \Pr(\epsilon_i < \beta \cdot X_i) \text{ (because the logistic distribution is symmetric)} \\ &= \text{logit}^{-1}(\beta \cdot X_i) = p_i \text{ (see above)} \end{aligned}$$

Logistic regression -- Part 6

Using the chi-squared test of significance, the integral of the chi-squared distribution with one degree of freedom from 11.6661... to infinity is equal to 0.00063649... This effectively means that about 6 out of a 10,000 fits to random y_k can be expected to have a better fit (smaller deviance) than the given y_k and so we can conclude that the inclusion of the x variable and data in the proposed model is a very significant improvement over the null model. In other words, we reject the null hypothesis with $1 - D \approx 99.94\%$ confidence.

Logistic regression -- Part 7

$$X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1m} \\ x_{21} & x_{22} & \dots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nm} \end{bmatrix}$$

Logistic regression -- Part 8

Wald statistic Alternatively, when assessing the contribution of individual predictors in a given model, one may examine the significance of the Wald statistic. The Wald statistic, analogous to the t-test in linear regression, is used to assess the significance of coefficients. The Wald statistic is the ratio of the square of the regression coefficient to the square of the standard error of the coefficient and is asymptotically distributed as a chi-square

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distribution.

Logistic regression -- Part 9

An intuition for this comes from the fact that, since we choose based on the maximum of two values, only their difference matters, not the exact values — and this effectively removes one degree of freedom. Another critical fact is that the difference of two type-1 extreme-value-distributed variables is a logistic distribution, i.e. $\varepsilon = \varepsilon_1 - \varepsilon_0 \sim \text{Logistic}(\mu, 1)$. $\varepsilon_1 - \varepsilon_0 \sim \text{Logistic}(\mu, 1)$. We can demonstrate the equivalent as follows:

Logistic regression -- Part 10

Fit The usual measure of goodness of fit for a logistic regression uses logistic loss (or log loss), the negative log-likelihood. For a given x_k and y_k , write $p_k = p(x_k)$. The p_k are the probabilities that the corresponding y_k will equal one, and $1 - p_k$ are the probabilities that they will be zero (see Bernoulli distribution). We wish to find the values of β_0 and β_1 which give the "best fit" to the data. For comparison of a "best fit" to the data, see the case of linear regression. There, the sum of the squared deviations of the fit from the data points (y_k) , the squared error loss, is taken as a measure of the goodness of fit, and the best fit is obtained when this loss is minimized. The log loss for the k -th point ℓ_k is: